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## Experiment No. 4

### Title

### Spectrophotometric Determination of Optical Band Gap of Cobalt (II) Chloride

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### Aim

To determine the optical band gap energy ( $E_g$ ) of Cobalt (II) chloride using UV–Visible spectrophotometry. (V-Lab)

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### Principle / Theory

When light passes through a material, photons with energy equal to or greater than the band gap can excite electrons from the **valence band** to the **conduction band**. This causes absorption at specific wavelengths.

The absorption coefficient ( $\alpha$ ) is related to photon energy ( $h\nu$ ) by the **Tauc equation**:  
Where:

- $\alpha$  = absorption coefficient
- $h\nu$  = photon energy (eV)
- $E_g$  = optical band gap
- $A$  = constant
- $n = \frac{1}{2}$  for **direct allowed transition**
- $n = 2$  for **indirect allowed transition**

For most metal chlorides, direct transition is assumed:

Plotting  **$(\alpha h\nu)^2$  vs  $h\nu$**  and extrapolating the linear portion to the x-axis gives  **$E_g$** .

### Importance of Cobalt(II) Chloride:

- Cobalt (II) chloride [ $\text{CoCl}_2$ ] is not a conventional semiconductor like silicon, but it plays an important role in the preparation and modification of semiconducting materials.
- $\text{CoCl}_2$  is widely used as a precursor for synthesizing cobalt-based semiconductors such as cobalt oxides, sulfides, and selenides, which are applied in photocatalysis, sensors, solar cells, batteries, and supercapacitors
- It serves as a dopant source for materials like  $\text{ZnO}$ ,  $\text{TiO}_2$ , and  $\text{CdS}$ , introducing new energy levels that improve conductivity, light absorption, and magnetic properties.
- Due to its strong electronic transitions,  $\text{CoCl}_2$  helps in “band gap engineering” and modification of optical properties.
- Being inexpensive, soluble, and easy to process,  $\text{CoCl}_2$  is suitable for large-scale and low-temperature synthesis.
- Thus, Cobalt (II) chloride is an important chemical for developing and tuning semiconducting materials.
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### Apparatus & Chemicals

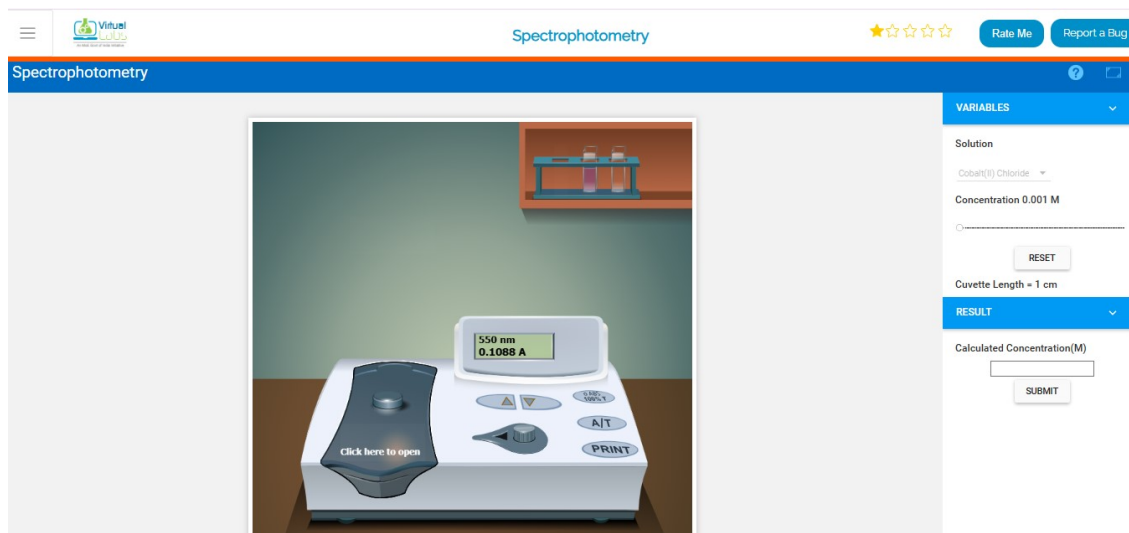
- UV–Visible spectrophotometer
- Quartz cuvettes
- Analytical balance
- Volumetric flasks
- Distilled water
- Cobalt (II) chloride ( $\text{CoCl}_2$ )
- Pipettes, beakers

NOTE: The virtual setup is available in simulation of physical chemical lab at amrita vlab portal.

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### Preparation of Sample Solution

1. Select Cobalt (II) chloride in from drop-down menu of solution.
2. Set concentration of Cobalt (II) chloride 0.001 M.



3. Click on “**Click to Open**” to open the UV–Visible spectrophotometer..
4. Place the **blank cuvette** in the sample holder and adjust the absorbance to zero. Remove the blank cuvette with a single click.
5. Place the cuvette containing **0.001 M Cobalt (II) chloride solution** in the sample holder.
6. Record the absorbance of 0.001 M Cobalt (II) chloride in the wavelength range **350–600 nm**.
7. Prepare an **Excel sheet** and enter the absorbance values at the corresponding wavelengths.
8. You can get approximate optical band gap values with formula,  
 $1240/\lambda_{\text{max}}$  .
9. Now for exact band gap measurement, we will make tauc plot with Use of **Origin software** (or Excel).
10. Calculate photon energy (**h $\nu$** ).
11. Calculate the **absorption coefficient ( $\alpha$ )** for each wavelength and ( **$\alpha h\nu$** )<sup>2</sup>.
12. Plot ( **$\alpha h\nu$** )<sup>2</sup> versus **h $\nu$** .
13. Extrapolate the straight-line portion of the plot to intersect the energy axis and determine the band gap energy.



## Calculations

### 1. Photon Energy:

$$h\nu = \frac{1240}{\lambda} \quad (eV)$$

Where  $\lambda$  is in nm.

### 2. Absorption Coefficient:

$$\alpha = \frac{2.303 \times A}{t}$$

Where:

- A = absorbance
- t = path length of cuvette ( $\approx 1$  cm)

### 3. Tauc Quantity:

$$(\alpha h\nu)^2$$

Plot against  $h\nu$ .

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## Graph

Plot:  $(\alpha h\nu)^2$  vs  $h\nu$

**Y-axis:**  $(\alpha h\nu)^2$

**X-axis:**  $h\nu$  (eV)

The straight line portion is extrapolated to intersect X-axis.

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**Add screen shot of the workbook and tauc plot from origin**

### **Result**

The optical band gap of Cobalt (II) chloride is found to be:

[ $E_g = \text{_____ eV}$ ]

(Typically reported value lies around **2.0 – 2.5 eV**, depending on form and conditions.)